

📁 PUJ-fungi repository · commit 28dad14

🧬 antiSMASH 8.0.4 + Funannotate

🛡️ re-audited 2026-09-08

Two Fungi, One Question.

Field Ally or Toxigenic Risk?

AF-PUJ (*Aspergillus flavus*) vs. TA-PUJ (*Trichoderma asperellum*) — agricultural biocontrol genomic review, AGRI-BIOCONTROL-GENOMIC-SCRUTINY V6

95

BGC regions inventoried & verified

4

Critical audit findings resolved in V6

~230

machine checks re-run on byte-verified data

The Two Isolates at a Glance

TA-PUJ · *Trichoderma asperellum*

● BIOCONTROL CANDIDATE

ASSEMBLY 20.29 Mb · 1,376 contigs (largest 62.7 kb) · GC 48.68%

GENES 5,229 CDS · 134 tRNA

BGC REGIONS 21 → 8 with MIBiG matches + 13 orphan

KEY TRAITS ACC deaminase (TA:PUJ_004816), 14 GH18 chitinases, 7 GH75 chitosanases, FET3-FTR1 iron-scavenging dyad

⚠ Draft assembly ≈55–60% of expected 33–40 Mb — absence claims are lower bounds, not proof (C1)

AF-PUJ · *Aspergillus flavus*

● TOXIGENIC — CLOSED-VESSEL USE ONLY

ASSEMBLY 36.79 Mb · 97 scaffolds (largest 2.40 Mb) · GC 48.15%

GENES 9,792 CDS · 259 tRNA

BGC REGIONS 74 → 34 with MIBiG matches + 40 orphan

KEY TRAITS complete PH0 phosphate regulon, 154 P450s, 24 GSTs, aerobactin-like NIS siderophore

⚠ 5 confirmed toxigenic BGCs — aflatoxin, CPA, aspirochlorine, aspergillilic acid, 3-NPA

V5 → V6 — What the Re-Audit Found

🟢 IN GOOD SHAPE

24/24

audit findings addressed in the V6 change log

95/95

BGC inventory rows re-verified exactly — lengths, scores, gene hits, identity ranges

95%+

of ~230 machine checks pass (Biopython + raw /db_xref parsing on byte-verified payloads)

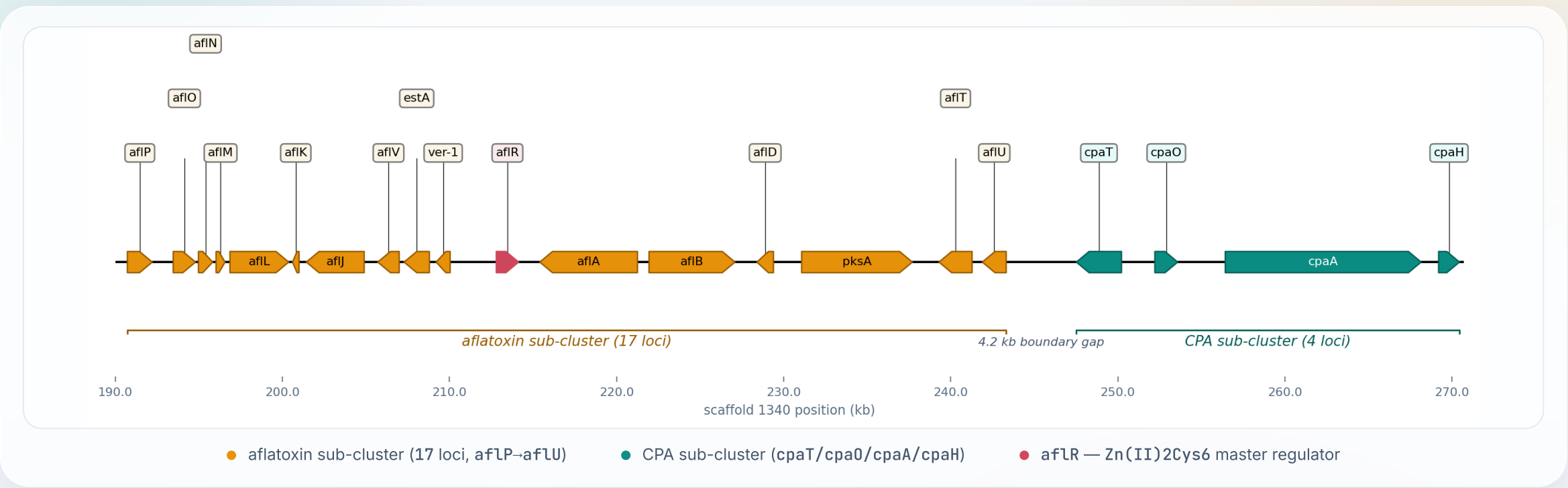
All four Critical findings are genuinely resolved

- ✓ C1 — TA assembly-completeness caveat added; absence claims downgraded to lower bounds
- ✓ C2 — TA mycotoxin verdict re-tiered: 13 orphan BGCs + incomplete assembly → LC-MS/MS screen mandated
- ✓ C3 — 5,098 cross-genome locus-tag collisions fixed via TA:/AF: prefixes throughout
- ✓ C4 — 3-nitropropanoic acid BGC (703_c2) added as 5th toxigenic cluster

⚠️ **Residual:** 4 Major + 8 Minor one-line errors (wrong locus tags / stale numbers) — none alters a conclusion. **Exact fixes on slide 8.**

The Key Gene Cluster — Aflatoxin + CPA Super-Cluster

Scaffold 1340 · 21 loci across 79.7 kb · rendered with dna_features_viewer (the plotting engine behind gbpars neighborhood --visualize)



94% / 97%

AflR / PksA identity to BGC0000007.3 (aflatoxin cluster, 11 genes, score 16,946)

4.2 kb

intergenic boundary between aflatoxin and CPA sub-clusters

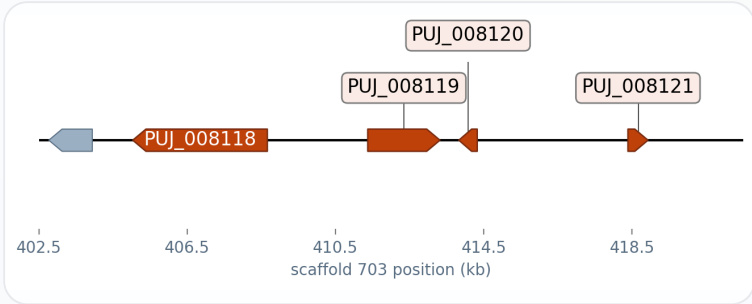
28–32 kb

deletion in commercial atoxigenic strains (Aflasafe, NRRL 21882) — **ABSENT in AF-PUJ**: the full cluster is present

Biosafety Gate — • Five Confirmed Toxigenic BGCs in AF-PUJ

TOXIN	MIBIG / RULE	GENES	SCORE	IDENTITY	
Aflatoxin B1/G1	BGC0000007.3	11	16,946	51-97%	IARC Group 1
Cyclopiazonic acid	BGC0000977.4	4	9,573	90-97%	Ca ²⁺ -ATPase inhibitor
Aspirochlorine (ETP)	BGC0001123.5	19	17,383	94-100%	thiol-reactive cytotoxin
Aspergillic acid	BGC0001516.5	6	6,227	75-100%	hepatotoxin
3-Nitropropanoic acid	rule (NpaA & NpaB)	4	region 17,956 bp	rule match	mitochondrial poison

• Newest addition — 3-NPA (C4)



Scaffold 703 · 403.9-421.8 kb · Puj_008118-008121 — missed in V5, now in the LC-MS/MS panel

⊘ AF-PUJ is **NOT** an atoxigenic biocontrol strain. No live field inoculation — closed-vessel fermentation or sterile bioremediation matrices only.

• TA-PUJ — Biocontrol Profile (and Its Caveats)

🌱 Plant growth & stress relief

- ✓ **ACC deaminase** Tas-acdS (TA:PUJ_004816) — EC 3.5.99.7, IPR005965-verified, the ethylene-rescue enzyme
- ⚠️ **IAA auxin synthesis:** T3 literature inference only — Salkowski + LC-MS/MS required (no YUCCA / ipdC found)

🧪 Secondary metabolites (21 BGC regions)

enniatin-like ionophore — contig_1710, 54% to BGC0000342.4

leucinostatin-like insecticidal depsipeptide — contig_1342, 52–86% to BGC0001358.4

metachelin siderophore — contig_1419 (scaffold-edge truncation)

8 of 21 regions have MIBiG matches

⚠️ 13 of 21 BGC regions are orphans — uncharacterized metabolic potential

🦠 Mycoparasitism arsenal

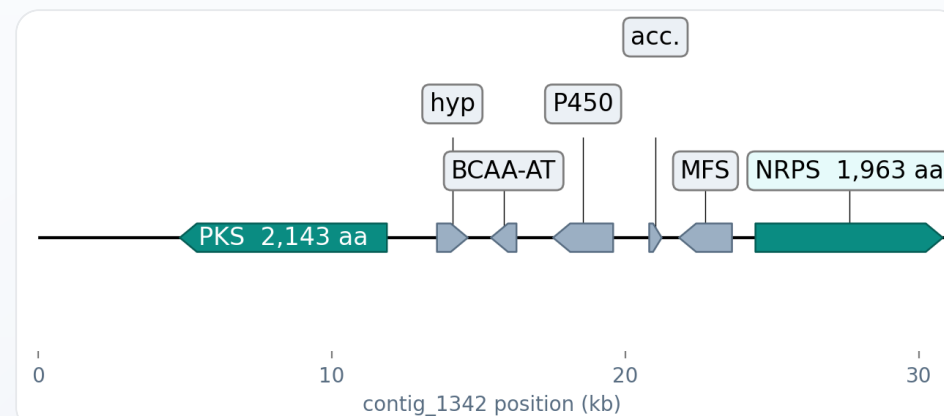
14 GH18 chitinases

7 GH75 chitosanases

16 S8 subtilisin proteases

FET3-FTR1 RIA dyad (contig_623)

Leucinostatin-like BGC · contig_1342



hybrid PKS (2,143 aa) + NRPS (1,963 aa) + P450 + MFS — classic insecticidal depsipeptide architecture

⚠️ Assembly ≈55–60% complete — no peptaibol megasynthetase ≥6,000 aa assembled, but collapse across 1,376 contigs cannot prove absence (Aib 85-Da neutral-loss MS/MS mandated)

Deployment Verdict

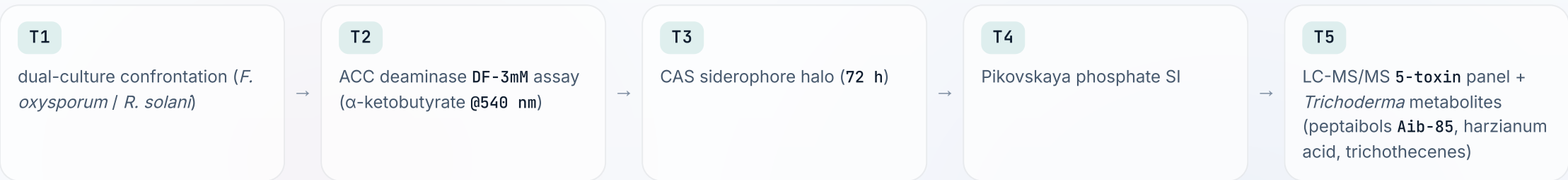
TA-PUJ → **field bio-inoculant** **CANDIDATE**

-  direct soil & seed inoculation, mycoparasitism + stress relief profile intact
-  gate: **Tier-5** metabolite screening before any field pilot
-  re-sequencing / long-read scaffolding + BUSCO (*hypocreales_odb10*) to close the completeness gap

AF-PUJ → **closed-vessel industrial strain** **ONLY**

-  enzyme fermentations (cellulases, amylases, phosphatases) with detoxified filtrate
-  sterile, non-food bioremediation matrices
-  **never live in open agricultural soils** — 5 toxigenic BGCs, intact aflatoxin/CPA cluster

5-tier wet-lab validation



Re-Audit Residuals — 12 One-Line Fixes Before V6.1

none alters a conclusion; all verified against the payloads

• 4 Major

R1 §8.2 nitrilase list — replace PUJ_000676/002829/003006/003664 with PUJ_001222 / 002797 / 003129 / 004143 (only 002797 carries EC 3.5.5.1)

R2 §1 BCAA row — AF:PUJ_002717 is Dicer-like dcl1; real EC 2.6.1.42 hits: TA 3 (003177/003357/004941), AF 4 (001094/001443/008295/009285)

R3 §2.B.6 — aspirochlorine GST is AF:PUJ_004908 (246 aa), not PUJ_004910 (422 aa dipeptidase)

R4 m4 — scaffold 1340 length is 280,200 bp, not 135,160

• 8 Minor

R5 region 1340_c1 end = 280,200 (§7B length 79,127 is right)

R6 §9.6 stale 23.9 Mb / 6,009 → 20.29 Mb / 5,229

R7 re-tier FET3 / PH05 / NIS ECs to T3 (not in GBK)

R8 m2 glucanase locus = TA:PUJ_002322, not 002476

R9 §8.1 AbT1 = BGC0000307.4 · 2,072 · 52% (as §3.A.7)

R10 glucanase products TA 8 / AF 4; AF phosphatase-by-product = 112

R11 §7 type column simplified; AF coords are region-local

R12 PUJ_000117 4-module claim is inference → T3

Trust the directory. Doubt the deletions. Screen before you sow.

● TA-PUJ

promising field candidate, pending metabolite
screen + re-assembly

● AF-PUJ

toxigenic; industrial closed-vessel value only

● V6

audit-verified core, 12 residual one-liners → **V6.1**

Data: AF gbk d3fb74bf367092e6 · 83,318,907 B — TA in c07f6c6e379fec38 · 45,029,346 B — TA antiSMASH b27db6f44d8fe3e7 · 44,880,695 B

Tools: Funannotate 1.8.17 · antiSMASH 8.0.4 · MIBiG · Biopython 1.84 · gbparse 0.8.1 + dna_features_viewer 3.1.5